Tested MedGemma 1.5 4B on 20 breast ultrasound images from the BUSI dataset. The images were selected using a fixed stratified split, consisting of 11 benign, 5 malignant and 4 normal images.

Dataset Splits: *busi_development_20.csv* (20 IMAGES) & *busi_test_760.csv* (REST 760 IMAGES)

Prompt: Used the same prompt in *busi_classification_v1.txt* that instructed MedGemma to classify each image and return its response in JSON format.

Results: *busi_development_20_medgemma.csv* & *busi_development_summary.csv*

Stats:

- **Total images tested:** 20
- **Valid JSON responses:** 20/20 (100%)
- **Correct classifications:** 5/20 (25%)
- **Incorrect classifications:** 15/20 (75%)
- **Predicted normal:** 9 images
- **Predicted benign lesion:** 4 images
- **Predicted malignant lesion:** 0 images
- **Predicted insufficient image quality:** 7 images
- **Average inference time:** 3.02 seconds per image

All 20 responses followed the required JSON format, demonstrating that MedGemma successfully understood and followed the structured output instructions. Only the JSON formatting was valid, but the results were still incorrect/missclassified.

Saved the above Output Schema as a CSV (*busi_development_20_medgemma.csv*)

I have added a few text cells in the 02_inference cell for my understanding of what each cell is doing, if I need to refer to them in the future.

Google Colab Link 02_inference.ipynb:
https://colab.research.google.com/drive/110Yui409o7Md7N1aD-6lini0PSddGKkl?usp=sharing